

HUMAN SAFETY DEVICE To Monitor Kids, Elderly, And Sick

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This human safety device offers a compact, reliable solution for detecting, monitoring, and responding to emergencies involving vulnerable individuals. Built using an IndusBoard Coin, it addresses the urgent need for immediate assistance, particularly for the elderly, children, and those with medical conditions.

The device integrates a MEMS sensor for fall detection, a temperature sensor for health monitoring, and GPS and GSM modules for location tracking and communication. Automatic and manual response capabilities enhance its effectiveness. The MEMS sensor identifies falls or sudden movements, sending alerts with GPS coordinates and sensor data to emergency contacts via SMS. The temperature sensor triggers alerts for abnormal readings, potentially indicating a health crisis. An emergency button allows manual alert activation, ensuring user control in critical situations.

The GSM module ensures alerts are delivered via SMS even in areas without internet connectivity, making it suitable for remote and outdoor locations. Despite dependencies on GSM network coverage and battery life, the device represents a significant advancement in personal

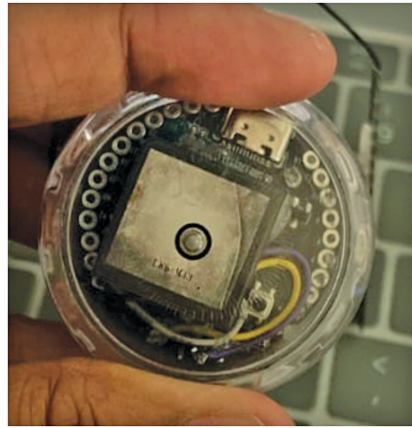


Fig. 1: Author's prototype

safety. Future iterations could incorporate additional sensors, wearable designs, and a mobile app for real-time tracking and control.

Harnessing advanced sensor and communication technologies, this device delivers a vital layer of security. Fig. 1 shows the prototype, Fig. 2 illustrates the block diagram, and required components are detailed in the Bill of Materials table.

Circuit diagram

Fig. 3 shows the circuit diagram of the human safety device. It is built around the IndusBoard Coin, L86 GPS, SIM800L, a 3V battery, a tactile switch, and a buzzer. Fig. 4 illustrates the pinouts for the IndusBoard, while Fig. 5 shows the pinouts for the L86 GPS module, which operates on the same serial USART as the GSM module.

ALERT,FALL AT LOCATION: <https://www.google.co.in/maps/place/17.52115,78.63043>

ALERT,EMERGENCY AT LOCATION: <https://www.google.co.in/maps/place/17.52114,78.63033>

ALERT,HIGH TEMPERATURE AT LOCATION: <https://www.google.co.in/maps/place/17.52112,78.63041>

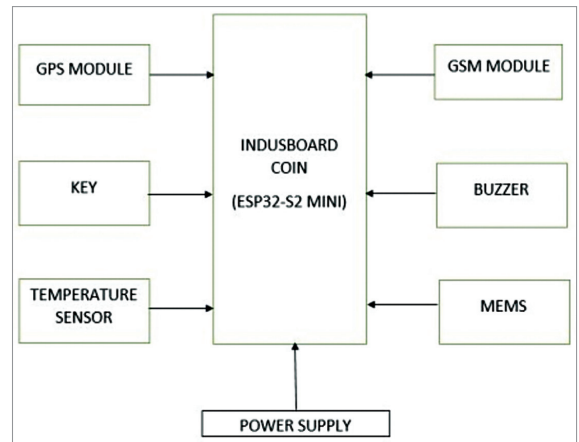


Fig. 2: Block diagram

The IndusBoard Coin is a compact IoT development board designed for innovative electronics devices, measuring only 3cm. It integrates temperature, magnetic compass, and acceleration sensors within the same 3cm size, along with over 10 capacitive touch sensors. Only the SIM800L and GPS modules need to be connected externally. The emergency button's built-in touch sensor can serve as a switch, but a physical switch is

BILL OF MATERIALS

Components	Quantity
IndusBoard Coin	1
SIM800L GSM module	1
L86 GPS module	1
Buzzer	1
Switch (optional)	1
3.3V or 3.7V battery	1